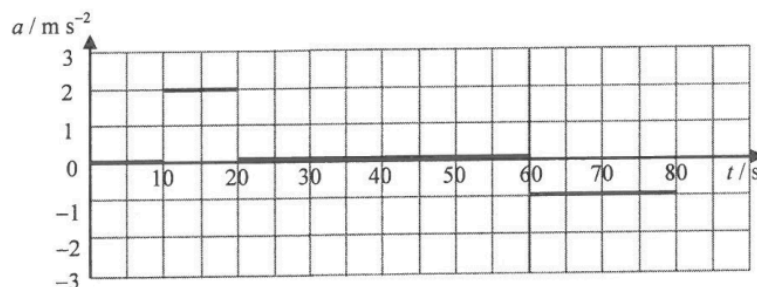


3. (a) 0 – 10 s: constant / uniform acceleration
10 – 80 s: constant velocity / uniform motion

1A
1A
2
1A
1A
2

- (b) (i) Car B.
It is represented by the steepest slope between 10 s and 20 s for B.
 $a = \frac{20 - 0}{20 - 10} = 2 \text{ m s}^{-2}$

(ii)



Uniform acceleration between 10 s and 20 s & uniform deceleration between 60 s and 80 s.

Or Either acceleration or deceleration correct.

All correct.

1A
1A
1A
2

- (c) (i) Total area of A by 20 s = $\frac{(10 + 20)}{2} \times 15 = 225 \text{ m}$
Total area of B by 20 s = $\frac{(10 \times 20)}{2} = 100 \text{ m}$
Separation of A & B, $s_{AB} = 225 \text{ m} - 100 \text{ m} = 125 \text{ m}$

1M
1A
2

- (ii) Let B meet A at time $(20 + T) \text{ s}$,
 $(v_B - v_A) \times T = s_{AB}$
 $(20 \text{ m s}^{-1} - 15 \text{ m s}^{-1}) \times T = 125 \text{ m}$
 $T = 25 \text{ s}$

$\therefore$ B catches up with A at $t = (20 + 25) \text{ s} = 45 \text{ s}$.

Or $s_A + v_A \times T = s_B + v_B \times T$
 $225 \text{ m} + 15 \text{ m s}^{-1} \times T = 100 \text{ m} + 20 \text{ m s}^{-1} \times T$
Or Accept using graphical method and read the answer from the graph

1A
1M
1M
1A
2

- (d) Let F_A and F_B be the driving forces developed by the engines of A and B respectively.
Ratio of power developed by cars A and B,
 $P_A : P_B = F_A \times v_A : F_B \times v_B$
 $= v_A^2 \times v_A : v_B^2 \times v_B$
 $= 3^3 : 4^3 = 27 : 64$

1M
1A
2

4. (a) The block decelerates uniformly (until at rest or its velocity is zero at $t = 1.5$ s).
It then moves with uniform acceleration down the plane (until $t = 3.5$ s).

1A

1A

2

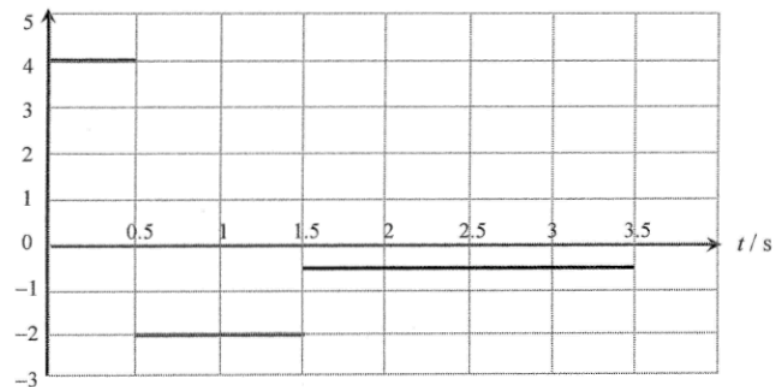
(b) (i) $a_2 = \frac{-1 - 0}{3.5 - 1.5}$
 $= 0.5 \text{ m s}^{-2}$

1M

1A

2

(ii)

 $a / \text{m s}^{-2}$ 

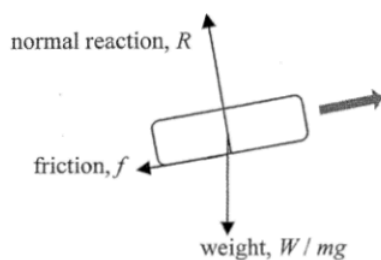
Either acceleration correct
All correct

1A

1A

2

(c)



Friction correctly indicated
All correct

1A

1A

2

- (d) Moving upward : $-mg \sin \theta - f = ma$
 $-(1)(9.81) \sin \theta - f = (1)(-2) \dots \dots \dots \textcircled{1}$

1M

- Moving downward : $-mg \sin \theta + f = ma'$
 $-(1)(9.81) \sin \theta + f = (1)(-0.5) \dots \dots \dots \textcircled{2}$

1M

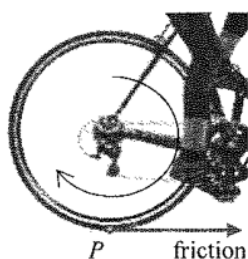
$\textcircled{2} - \textcircled{1} : 2f = 1.5$
 $f = 0.75 \text{ N}$
(Note: $\theta = 7.32^\circ$)

1A

3

3. (a) $a = \frac{6-0}{2-0}$ $= 3 \text{ m s}^{-2}$ (downwards)	1M 1A	
(b) A: 395 N B: 569 N C: 685 N In stage B, balance reading = weight (Newton's 1 st law), $mg = m \times 9.81 \text{ m s}^{-2} = 569 \text{ N}$ $m = 58.0 \text{ kg}$ Or In stage A, according to Newton's 2nd law $(569 - 395) \text{ N} = ma = m(3 \text{ m s}^{-2})$ $m = 58.0 \text{ kg}$	2 1A 1M 1A 1M 1A	For $g = 10 \text{ m s}^{-2}$, $m = 56.9 \text{ kg}$
(c) (i) For stage C, by Newton's 2 nd law, $F = ma$ $(569 - 685) \text{ N} = (58.0 \text{ kg}) a$ $a = -2 \text{ m s}^{-2}$ Thus, $a = \frac{0-6}{T-12} = -2 \text{ m s}^{-2}$ $T = 15 \text{ s}$	3 1M 1M	For $g = 10 \text{ m s}^{-2}$, $m = 56.9 \text{ kg}$, $a = -2.04 \text{ m s}^{-2}$
(ii) Height $\approx$ displacement of lift = area under graph $= \frac{(12-2)+15}{2} \times 6$ $= 75 \text{ m}$	2 1M 1A 2	

3. (a)



1A

1

- (b) Mechanical power delivered (for work done against the resistive force)

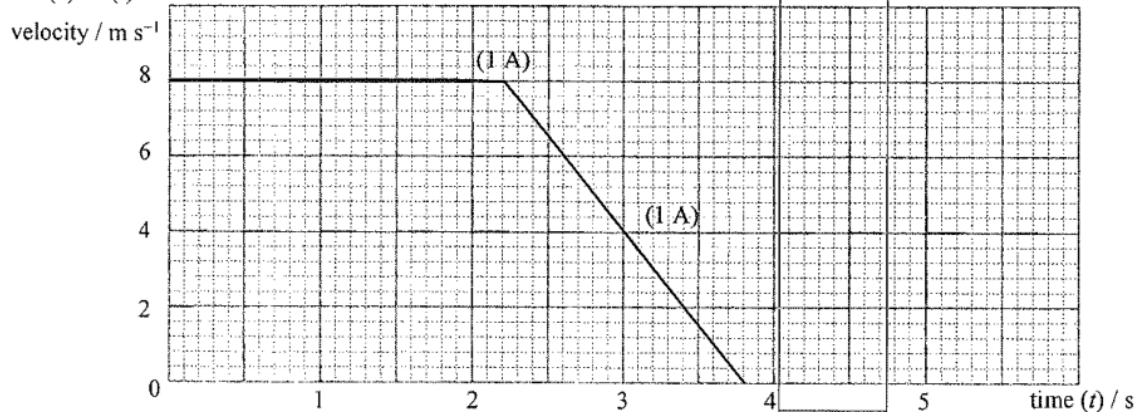
$$\begin{aligned}
 P &= Fv \\
 &= 17.0 \times 8.0 \\
 &= 136 \text{ W}
 \end{aligned}$$

1M

1A

2

- (c) (i)



2

- (ii) Deceleration: $\left| \frac{0-8}{3.8-2.2} \right| = 5.0 \text{ m s}^{-2}$
 Resistive force = 65×5.0
 $= 325 \text{ N}$

1M

e.c.f. from (c)(i)

1A

Alternative method:

$$\text{Braking distance} = \frac{1}{2}(8)(3.8 - 2.2) = 6.4 \text{ m}$$

$$Fs = \frac{1}{2}mv^2 - 0$$

$$F(6.4) = \frac{1}{2}(65)(8)^2$$

$$F = 325 \text{ N}$$

1M

1A

2

- (iii) Total distance travelled after $t = 2.0 \text{ s}$
 $= \text{area under the graph from } t = 2.0 \text{ s to } t = 3.8 \text{ s}$
 $= \frac{0.2+1.8}{2} \times 8.0 = 8.0 \text{ m}$
 Distance between the bicycle and the obstacle when the bicycle stops = $9 - 8$
 $= 1 \text{ m}$

1M/1A

1M

1A

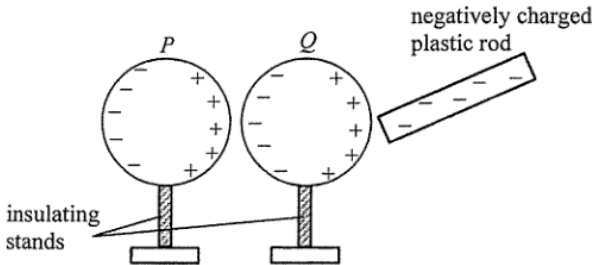
3

- (d) The padded cushion can increase the contact time of the knee and the ground such that a smaller force acts on the knee.

1A

1A

2

9. (a)		2A		
			2	
(b) No, it remains unchanged as charging is through induction (without any contact or transfer of charges).	1A+1A		2	
(c) Hold / bring the unknown charged rod (assume positive) near sphere P, which carries positive charges, without touching it.	1A			
If P is repelled by the rod, the rod should carry positive charges. <u>And / Or</u> If P is attracted by the rod, the rod should carry negative charges.	1M	1M for the correct procedure to identify the polarity of the charged rod.		
			2	

3. (a) The object is moving to the left with constant deceleration and (momentarily) stops at $t = 4$ s.

(b) Displacement = total area of the graph

$$= \frac{1}{2} \times (-10) \times 4 + \frac{1}{2} \times 15 \times (10 - 4)$$

$$= 25 \text{ m}$$

Position of the object is 25 m from point O on the right side.

(c) Net force acting on the object

$$= ma$$

$$= 0.4 \times \frac{15 - (-10)}{10 - 0}$$

$$= 1 \text{ N (towards right)}$$

1A	2
1M	2
1A	2
1M	2
1A	2

Or

$$v^2 = u^2 + 2as$$

$$15^2 = (-10)^2 + 2 \times \frac{15 - (-10)}{10 - 0} \times s$$

$$s = 25 \text{ m}$$